

Simulator for networks lab, projects and research

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TM

Netsim

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protocol model libraries with source code

PSpice has analogue and digital libraries of standard components such as NAND, NOR, flip-flops and other digital gates, op-amps, etc. This makes it a useful tool for a wide range of analogue and digital applications.

VHDL is used in electronic design automation to describe digital and mixed-signal systems, such as field-programmable gate arrays and integrated circuits. It is commonly used to write text models that describe a logic circuit. Such a model is processed by a synthesis program, only if it is part of the logic design.

A simulation program is used to test the logic design using simulation models to represent the logic circuits that interface the design. This collection of simulation models is commonly called a testbench. The key advantage of VHDL when used for system design is that it allows the behaviour of the required system to be described (modelled) and verified (simulated) before synthesis tools translate the design into real hardware (gates and wires).

Q3. Please explain the following:

1. Maximum power-point tracking (MPPT) charge controller
2. Difference between copper-indium gallium-selenide (CIGS) and crystalline solar panels

Avijit Dasgupta
Through e-mail

A3. **MPPT.** It is the algorithm used for extracting maximum available power from a photovoltaic (PV) module by making it operate at the most efficient voltage (maximum power point). MPPT checks the output of the PV module, compares it with the battery voltage and calculates the best power that the PV module can produce to charge the battery. It then converts it into the best voltage to get maximum current into the battery. It can also supply power to a DC load, which is connected directly to the battery.

MPPT is most effective in cold weather, cloudy or hazy days. Normally, PV modules work better at cold temperatures and MPPT is utilised

to extract maximum power available from these. It is also effective when the battery is deeply discharged; MPPT can extract more current and charge the battery if the charge in the battery is low.

Main features of an MPPT solar charge controller are:

1. It allows the solar power system to extract maximum power from the PV module. It forces the PV module to operate at a voltage close to the maximum power point to draw maximum available power.

2. It allows use of PV modules with a higher voltage output than the operating voltage of the battery system. For example, if a PV module has to be placed far away from the charge controller and battery, its wire size must be very large to reduce the voltage drop. With an MPPT solar charge controller, users can wire a PV module of 24V or 48V (depending on the charge controller and PV modules) and bring power into a 12V or 24V battery system. This means that the MPPT solar charge controller reduces the wire size needed while retaining the full output of the PV module.

Difference between CIGS and crystalline solar panels. Bulk of the solar cells is made from crystalline silicon, which can be monocrystalline, polycrystalline, multicrystalline or rubber silicon. There are also thin-film solar cells, which are made from CIGS and other materials. The active layer (CIGS) can be deposited in a polycrystalline form directly onto molybdenum-coated glass sheets or steel bands. This uses lesser energy than growing large crystals, which is a necessary step in the manufacture of crystalline-silicon solar cells. Also, unlike crystalline silicon, these substrates can be flexible.

Like all thin-film solar panels, CIGS panels are not as efficient as crystalline silicon solar cells. These are, however, most efficient of thin-film technologies.

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